ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 2 SET SPECIAL I **WINTER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 20**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

---

1. Given, 2 + 2 (CO2) (PO1)

$$S = \{\begin{bmatrix} x^4 \\ y^{12} \end{bmatrix} \mid \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2\}$$

This set represents all vectors that are scaled by a factor of 2 from the original vector space $\mathbb{R}^2$. **Is S a vector space?** If yes, justify your answer. If no, provide a counterexample to demonstrate why it is not.

2. 16 (CO2) (PO1)

Given,

$$A = \begin{bmatrix} 1 & 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 & 3 \\ 1 & 1 & 3 & 1 & 4 \\ 1 & 2 & 5 & 1 & 5 \end{bmatrix}$$

Answer the following questions (After reducing it to its Reduced Row Echelon Form (RREF)):

(a) What is the dimension of Matrix A?

(b) What is the rank of Matrix A?

(c) Identify the pivot and free columns of Matrix A.

(d) For all the four fundamental subspaces associated with Matrix A (Columnspace $C(A)$, Nullspace $N(A)$, Rowspace $C(A^T)$, Left Nullspace $N(A^T)$), state the following:

- Their dimensions.
- Bases of the subspaces.
- The subspaces they belong to (i.e., $\mathbb{R}^m$ or $\mathbb{R}^n$, or something else).
- Which two pairs are orthogonal to each other and justify by showing the dot product of their basis vectors.